



HUZEFAI
EMPOWERING THE SUCCESS WITH AI

Seminar on Generative AI

Offline Class @ Nungambakkam, Chennai

Date : 4th May 2025

Day: Sunday

Time: 9:30 AM-12:30 PM

Topics Covered:

GenAI, GenAI Models

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Limited Seats only | Register before you come

What is Generative AI

Generative Artificial Intelligence, often called gen AI.

It is a type of AI that can create new content like conversations, stories, images, videos, and music.

It can learn about different topics such as languages, Programming, art, science, and more, and use this knowledge to solve new problems.

Use of Gen AI

It can learn about popular design styles and create a unique logo for a brand or an organization.

Businesses can use Gen AI in many ways, like building chatbots, creating media, designing products, and coming up with new ideas.

Evolution of Gen AI

Generative AI has come a long way from its early beginnings.

Lets see how it has evolved
over the time

The Early Days: Rule-Based Systems

AI systems followed strict rules written by humans to produce results.

These systems could only do what they were programmed for and couldn't learn or adapt.

For Ex: A program could create simple shapes but couldn't draw something creative like a landscape.

Introduction of Machine Learning

(1990s-2000s)

AI started using Machine Learning. Which allowed it to learn from data instead of just following rules.

The AI was fed large datasets (Eg: Pictures of Animals), and it learned to identify patterns and make predictions

Eg: AI could now recognize a dog in a picture, but it still couldn't create a picture of a dog on its own.

The Rise of Deep Learning (2010s)

Deep Learning improved AI significantly by using neural networks, which mimic how the human brain works.

AI could now process much more complex data, like thousands of photos, and start generating new content.

Eg: AI could now create a realistic drawing of a dog by learning from millions of dog photos.

Generative Adversarial Networks (2014)

GANs, introduced in 2014, use two AI systems that work together: one generates new content, and the other checks if it looks real.

This made Gen AI much better at creating realistic images, videos and sounds.

Eg: GANs can create life like images of people who don't exist or filters. (FaceApp or Snapchat)

Large Language Models and Beyond (2020)

Models like GPT-3 and GPT-4 can understand and generate human-like text.

They are trained on massive amounts of data from books, websites, and other sources.

AI can now hold conversations, write essays, generate code, and much more..

Eg: ChatGPT can help you draft an email, write a poem, or even solve problems.

Multimodal Gen AI (Present)

New AI models can handle multiple types of data at once — text, images, audio and video.

This allows AI to create content that combines different formats.

Eg: AI can take a written description and turn it into an animated video or a song with the help of different models integration together.

Types of Generative AI Models

Gen AI is versatile, with different models designed for specific tasks. Here are some types.

Text-to-Text

Text-to-Audio

Text-to-Image

Text-to-Video

Image-to-Text

Multimodal AI

Speech-to-Text

Text-to-Text

These models generate meaningful and coherent text based on input text.

They are widely used for tasks like drafting emails, summarizing lengthy documents, translating languages or even writing creative content.

Tools like ChatGPT is brilliant at understanding context and producing human-like responses.

Text-to-Image

This involves generating realistic images from descriptive text.

Eg: Tools like DALL-E 2 can create a custom digital image based on prompts such as “A peaceful beach with palm trees during a beautiful sunset”

Offering endless possibilities for designers, artists, and marketers.

Image-to-Image

These models enhance or transform images based on input image.

Eg: they can convert a daytime photo into a night time scene, apply artistic filters, or refine low-resolution images into high-quality visuals.

Speech-to-Text

This application converts spoken words into written text.

It powers virtual assistants like siri, transcription software, and automated subtitles, making it a vital tool for communication, accessibility, and documentation.

Text-to-Audio

Generative AI can create music, sound effects, or audio narrations from textual prompts.

This empowers creators to explore new soundscapes and compose unique auditory experiences tailored to specific themes or moods.

Text-to-Video

These models allow users to generate video content by describing their ideas in text.

Eg: A marketer could input a vision for a promotional video and the AI generates visuals and animations, streamlining content creation.

Multimodal AI

These systems integrate multiple input and output formats, like text, images, and audio, into a unified interface.

Eg: An educational platform could let students ask questions via text and receive answers as interactive visuals or audio explanations, enhancing learning experiences.

Benefits of Generative AI

Gen AI offers innovative tools that enhance creativity, efficiency, and personalization across various fields.

Enhances Creativity

Drives Economic Growth

Accelerates Research and Development

Improves Personalization

Empowers Non-Experts

Enhances Creativity

Gen AI enables the creation of original content like images, music and text, helping artists, designers, and writers explore fresh ideas.

Accelerates Research and Development

In fields like science and technology, Gen AI reduces the time needed for research by generating multiple outcomes and predictions.

Improves Personalization

Gen AI creates tailored content based on user preferences. From personalized product designs to customized marketing campaigns.

Empowers Non-Experts

Even users without expertise can create high-quality content using Gen AI.

This helps individuals learn new skills, access creative tools, and open doors to personal and professional growth.

Limitations of Generative AI

While Gen AI offers many benefits, it also comes with certain limitations that need to be addressed.

Data Dependence

Limited Control Over Outputs

High Computational Requirements

Ethical and Legal Concerns

Data Dependence

The accuracy and quality of Gen AI outputs depend entirely on the data it is trained on.

Limited Control Over Outputs

Gen AI can produce unexpected or irrelevant results, making it challenging to control the content and ensure it aligns with specific user requirements.

High Computational Requirements

Training and running Gen AI models demand significant computing power.

Ethical and Legal Concerns

Gen AI can be misused to create harmful content, like deep fakes or fake news, which can spread misinformation or violate privacy.